Choosing a Vision Tower for a Text-Only LLM

A frozen-representation study of six vision towers

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Abstract

Vision encoders with similar recognition accuracy can support different spatial tasks. We compare six frozen vision towers across recognition, depth, surface normals, correspondence, and occlusion, and examine internal layers and native multimodal connectors. Three patterns emerge. Capability profiles differ across towers and label budgets; the two highest full-label recognition estimates are statistically unresolved. Geometry peaks before the final layer in 23 of 24 task–tower–capacity combinations, while recognition peaks at the final layer in 19 of 24 tower–readout–capacity combinations. Finally, projected features improve probe performance over final tower features in all 16 DIODE geometry comparisons, with 15 paired intervals excluding zero. This pattern does not extend to KITTI, where two comparisons reverse and two are unresolved, or uniformly across correspondence datasets. These are measurements of recoverability under specified readouts, not information content or downstream language-model performance. A lossless merge changes correspondence scores substantially, illustrating that distinction. We document the unresolved recognition-pooling control, native-grid confound, and protocol corrections alongside the results.

1 Introduction

Adding vision to a language model requires choosing an encoder, an extraction layer, and a connector. A full vision-language model (VLM) evaluation measures their combined behavior after training. Frozen-feature probes answer a narrower question: what can a specified readout recover at each point in the visual pipeline? This can identify candidate representations before multimodal training, although their usefulness to a language model must still be tested downstream.

We study DINOv2 [18] and SigLIP2 [23] as standalone references, alongside towers extracted from Kimi K2.6 [17], Kimi K3 [12], Muse Glimmer [15], and Qwen3.8 [19]. Recognition asks which category an image contains. Geometry asks how far surfaces are from the camera and which direction they face. Correspondence asks whether features match the same point across views or related parts across objects. Occlusion tests recognition as visible image content is removed. These capabilities need not favor the same representation.

The benchmark resolves internal layers and, for multimodal towers, the native merge and projector stages. Towers and connectors remain frozen; recognition and geometry use trained readouts, while correspondence uses direct feature matching. The study contributes capability profiles for six towers, layer sweeps under two readout-capacity settings, and paired evidence that connector effects change across evaluation datasets. The accompanying code and result records expose the measurements and the controls behind these claims. We report effects under this protocol rather than proposing an overall tower ranking.

Table 1: Tower architectures. Tokens are measured before merging at 448×448. The last column is the projector's output width. DINOv2 and SigLIP2 have no native connector in this study.

Tower	Layers	Width	Patch	Tokens	Projected width
DINOv2-L	24	1024	14	1024	—
SigLIP2-so400m	27	1152	14	1024	—
Kimi K2.6	27	1152	14	1024	7168
Kimi K3	27	1024	14	1024	7168
Muse Glimmer	50	1536	14	1024	6656
Qwen3.8	27	1152	16	784	5120

2 Related work

Prismatic VLMs and Cambrian-1 compare visual encoders through multimodal training and complete VLM evaluations [11, 22]. Our study complements that approach by stopping at frozen visual features. Probe3D provides the closest methodological foundation: frozen-feature probes for depth and normals, and feature matching for correspondence [6]. We adapt its dense readout to one extraction layer at a time and use DIODE for geometry. These changes support layer comparisons but prevent interpreting our scores as a reproduction of its published benchmark.

Intermediate features can outperform final features. Perception Encoder demonstrates this across downstream tasks and introduces alignment methods to expose those features at the output [2]. We compare recognition and geometry along normalized layer depth and follow the deepest features through native connectors. Work beyond the encoder also motivates this stage analysis. Fu et al. [7] find visual information accessible inside VLMs that nevertheless perform poorly on vision-centric tasks. Wu et al. [26] find a drop in segmentation decodability at the adapter, followed by recovery in the language model. These studies use different tasks and readouts; they do not establish a universal preservation or loss rule. Our contribution is a joint comparison of towers, layers, and connector stages under a shared protocol, with the limitations of probing made explicit [1].

3 Benchmark design

3.1 Towers and extraction points

Table 1 lists the six towers. Each receives an image resized and center-cropped to 448×448, with its native normalization and input packing. At layer i of a tower with L blocks, relative depth is i/L. We sample $i = \text{round}(Lk/8)$ for $k = 1, \ldots, 8$; actual fractions vary slightly with L. The deepest extraction includes final tower normalization where present; Qwen3.8 is read at the last block output, before normalization on the merger path.

For each multimodal tower, we also extract `merged`, which concatenates each 2×2 group of patch tokens, and `projected`, the learned map into the parent language model's embedding width. These stages exist only after the final tower layer. We do not apply projectors to intermediate layers or exchange them between towers. Thus the experiment separates extraction points within native pipelines, rather than independently crossing every tower, layer, and projector.

3.2 Tasks and readouts

Recognition uses the 13,000-image validation export of `ilee0022/ImageNet100`, a 100-class ImageNet subset [4, 9], split by us into 9,100 training, 1,950 validation, and 1,950 test images. A linear classifier reads either the mean feature or a learned attention pool. Both operate on a cached 4×4 pooled grid. We report top-1 accuracy after 20 training epochs. Label-budget runs keep validation and test fixed and train on 100, 455, 1,820, or 9,100 images. The nominal 1% condition uses one image per class, hence 1.10% of the training split.

Geometry uses DIODE depth and normals [25], with 541 training, 115 validation, and 115 test images drawn from its validation release. The test split contains 51 indoor and 64 outdoor images. A Probe3D-style multiscale decoder reads the full patch grid from one layer and trains for 10 epochs. Depth is summarized by δ_1, the mean per-image fraction of valid pixels satisfying $\max(\hat{d}/d, d/\hat{d}) < 1.25$, for prediction $\hat{d}$ and target d; higher is better. Normals use mean angular error in degrees; lower is better. Targets cover the same crop as the input image.

Correspondence uses cosine feature matching on NAVI, ScanNet, and SPair-71k [3, 10, 16]. NAVI measures recall within 2 cm across views of an object; ScanNet measures recall within 10 pixels across indoor views; SPair measures keypoint accuracy within 10% of the target bounding-box scale across object instances. NAVI and ScanNet match on a 64×64 feature grid; SPair matches annotated keypoints against each stage's own grid. These metrics are not averaged together. Occlusion measures top-1 accuracy after masking 0, 10, 20, 35, or 50% of image area, with readouts trained once on clean images.

KITTI provides a separate depth experiment with 700 training, 150 validation, and 150 test images [8, 24]. We retrain the depth readout on KITTI to test whether the DIODE stage ordering recurs. This is not zero-shot transfer of a DIODE-trained readout. Dataset preparation and evaluation details appear in Appendix A.

3.3 Capacity, selection, and uncertainty

The core recognition and DIODE sweeps run at native feature width (*raw*) and after training-split PCA to 512 dimensions (*matched*). PCA is frozen before readout training. Matching holds head parameter count constant within a readout type; it does not equate spatial sampling or the information retained by the reduction. We retain both arms as a sensitivity analysis.

A cell specifies tower, extraction point, task/readout, and capacity arm. The core sweeps contain 224 recognition and 224 geometry cells. Each selects its learning rate on validation from a shared task-specific grid, then trains with three seeds. Headline intervals use 10,000 paired bootstrap resamples at 95% confidence. For trained readouts, we first average each image's score over seeds, then resample images; correspondence resamples image pairs. These intervals condition on the fitted runs and do not include all training uncertainty. We report seed standard deviations separately. Intervals are pointwise, without a multiple-comparison correction. An interval crossing zero is unresolved, not evidence of equivalence; an untested pair has no inferential ranking.

3.4 Controls and corrections

Extraction tests compare weights with parent checkpoints and restore spatial token order before dense evaluation. Kimi K3 has tower-weight equality but lacks the forward-output and projector-weight checks available for the other extracted models. Lossless merging is verified against cached tensors for the two Kimi towers and inferred from architecture for Muse Glimmer and Qwen3.8.

The pooled recognition cache has an incomplete control. On SigLIP2 and Muse Glimmer, mean-readout results agree with full-token features, but six of sixteen full-token attention cells

Table 2: Capability profiles. Recognition uses the label-budget runs at 1% and 100%, with final-layer features and the matched attention readout. Depth and normals are the best observed matched test scores across sampled layers, not independently validated layer selections. NAVI uses the final layer. Retention is accuracy at 50% occlusion divided by clean accuracy. Higher is better except for normals. No overall rank is assigned.

Tower	1% labels	Full labels	Depth ↑	Normals ↓	NAVI ↑	Retention
DINOv2	0.3689	0.9079	0.7015	18.85	0.5389	0.5194
SigLIP2	0.6559	0.9168	0.6702	23.90	0.4013	0.4883
Muse Glimmer	0.4330	0.9212	0.6588	24.30	0.2139	0.5163
Kimi K2.6	0.3448	0.8862	0.6595	25.51	0.3646	0.2786
Qwen3.8	0.2451	0.8791	0.6736	24.24	0.3785	0.3046
Kimi K3	0.2191	0.8369	0.6546	27.03	0.3334	0.2042

collapse during optimization. Follow-up runs do not establish stable agreement. Attention-based recognition levels and layer curves therefore remain conditional on pooled features.

Two corrections changed the recorded results: expanding the recognition learning-rate grid and seeding randomized PCA. Initially, 107 of 224 cells selected the lower rate limit. In the corrected run none selects the lower limit and three select the upper limit. All matched recognition cells and headline geometry comparisons were rerun. Together these corrections changed an apparent Muse Glimmer merge-to-projector loss from -0.0075 to $+0.0042$ top-1. The full matched geometry sweep retains its original randomized reductions. Its curves and selected peaks are descriptive; geometry intervals refer to deterministic reruns. Appendix B records the corrections.

4 Results

4.1 Capabilities and label budgets favor different towers

Table 2 shows capability profiles without an aggregate score. DINOv2 has the highest measured depth, normal, and NAVI performance, while Muse Glimmer has the highest full-label recognition estimate and the lowest tower-stage scores on all three correspondence datasets. These are observed levels; most cross-tower pairs have not been tested with intervals.

Three full-label recognition comparisons have paired intervals in the matched attention arm. Muse Glimmer exceeds DINOv2 by 0.0125 $[0.0034, 0.0215]$ top-1. Muse Glimmer versus SigLIP2, 0.0041 $[-0.0036, 0.0121]$, and SigLIP2 versus DINOv2, 0.0084 $[-0.0009, 0.0178]$, remain unresolved. The data therefore do not establish a unique full-label recognition winner.

The label budget changes the observed ordering (Figure 1). With one training image per class, SigLIP2 reaches 0.6559 top-1, compared with Muse Glimmer's 0.4330. At full labels their estimates converge to 0.9168 and 0.9212. Qwen3.8's nominal 1% cell is unstable: its seed standard deviation is 0.0780 around a mean of 0.2451. These curves describe label efficiency under the pooled attention readout, whose full-token control remains inconclusive.

Two cross-tower geometry comparisons were rerun with deterministic PCA at layers selected from the test-score sweep, without independent validation of that selection. DINOv2 exceeds Qwen3.8 in depth by 0.0292 $[0.0099, 0.0484]$ δ_1, but that comparison also changes native grid size from 28×28 to 32×32. DINOv2 reduces normal error relative to SigLIP2 by 5.0370 $[4.3038, 5.8164]$ degrees; both use 32×32 grids. These intervals support the named comparisons under their protocols, not a complete geometry ranking.

Occlusion adds another distinction. At 50% masking, DINOv2, Muse Glimmer, and SigLIP2

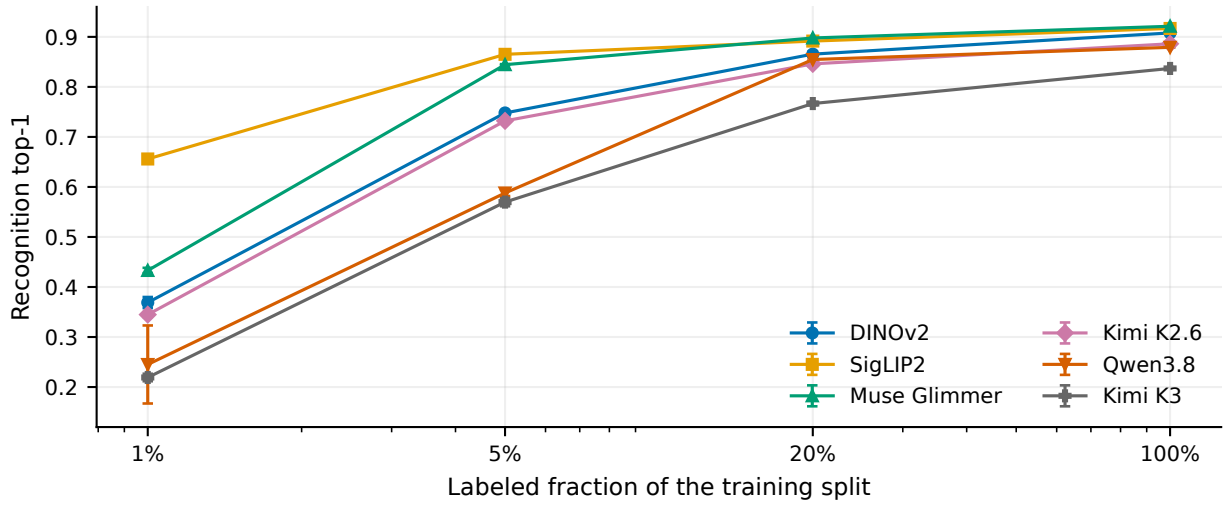


Figure 1: Recognition across label budgets at the final tower layer, matched attention arm. Points average three seeds; vertical bars show one seed standard deviation, not confidence intervals. The nominal 1% budget contains 100 images.

retain approximately half their clean accuracy, compared with 20–31% for the other three towers. This grouping is descriptive: available occlusion intervals compare clean and perturbed images within a stage, not retention between towers. Appendix E gives the curves and stage-level interpretation.

4.2 Geometry usually peaks before the final layer

Across two capacity arms, geometry has a pre-final test-score peak in 23 of 24 combinations of tower, task, and capacity arm. The exception is DINOv2 depth in the raw arm. Recognition reaches its highest score at the final layer in 19 of 24 combinations of tower, readout, and arm. Figure 2 shows attention recognition, depth, and normals for both arms; Appendix C lists the geometry peaks.

In the matched arm, choosing the final layer instead of the best observed layer reduces depth δ_1 by 0.0256–0.1422 and increases normal error by 4.34–10.22 degrees. These are descriptive sweep differences: the peak was identified on the same test scores being compared, and intervals were not computed for every peak-to-final contrast. The result motivates validating intermediate layers for spatial tasks. It does not establish a universal optimal fraction of tower depth.

Qwen3.8 accounts for four recognition curves whose peak is earlier than the final layer; Kimi K2.6 raw attention accounts for the fifth. Qwen3.8's earlier-layer advantage resolves in raw attention, 0.0128 $[0.0053, 0.0203]$, and raw mean, 0.0156 $[0.0053, 0.0263]$. Its two matched comparisons cross zero. The nominal Kimi K2.6 decline is smaller than its seed standard deviation and was not tested with an interval. Thus even this exception depends on the readout and capacity setting.

4.3 Connector effects change across evaluations

We expected the connector to reduce spatial recoverability. DIODE gives the opposite pattern: all 16 projected-versus-final-tower comparisons favor projected features, and 15 paired intervals exclude zero. These comprise four connectors, two geometry tasks, and two capacity arms. The unresolved comparison is Kimi K2.6 raw depth, 0.0196 $[-0.0003, 0.0402]$ δ_1.

The matched DIODE gains range from 0.0174 to 0.0784 in depth δ_1 and from 0.7924 to 4.3130 degrees of normal-error reduction. The largest raw depth gain is Qwen3.8's 0.1601, compared with

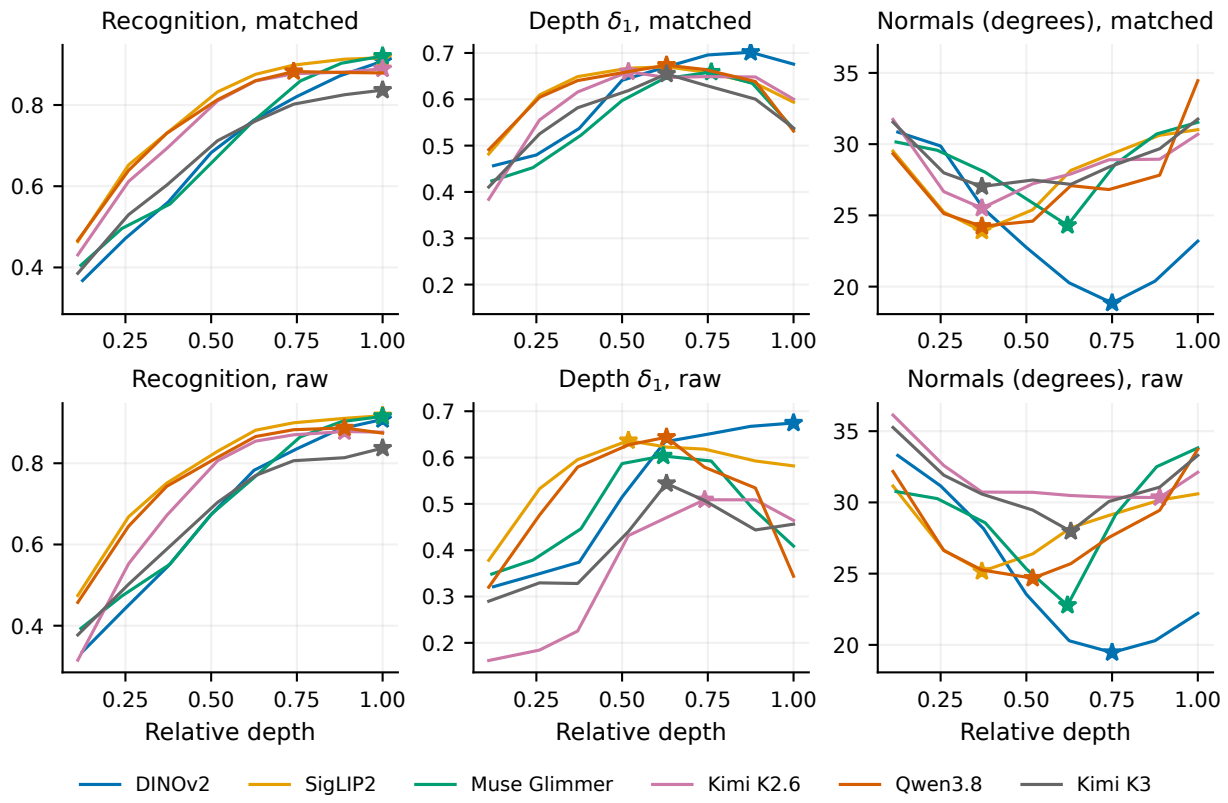


Figure 2: Layer sweeps at the tower stage. Columns show recognition with attention pooling, DIODE depth, and DIODE normals; rows show matched and raw arms. Stars mark each curve's best observed test score. Recognition uses pooled tokens; geometry uses full patch grids. Matched geometry curves come from the original randomized-PCA sweep.

0.0584 after capacity matching. Because PCA also changes the input representation, the difference between arms is a sensitivity result rather than an isolated estimate of the effect of head size.

On KITTI, Kimi K2.6 and Muse Glimmer reverse: projected-minus-tower differences are -0.0048 $[-0.0077, -0.0016]$ and -0.0039 $[-0.0074, -0.0007]$ δ_1, respectively. Kimi K3 and Qwen3.8 are unresolved. Training new readouts on driving scenes therefore does not reproduce the uniformly positive DIODE pattern. The datasets differ in scene content, target coverage, depth range, and sample size; this study does not isolate which difference causes the reversal.

Correspondence yields ten positive and two negative intervals, all excluding zero. Qwen3.8's projected-minus-tower difference is $+0.0188$ $[0.0146, 0.0230]$ on NAVI but -0.0194 $[-0.0208, -0.0180]$ on ScanNet. Kimi K3's SPair difference is -0.0144 $[-0.0216, -0.0070]$, while both geometric correspondence differences are positive. SPair compares parts across different object instances, so this result concerns a different capability from NAVI's same-object multiview consistency.

4.4 Lossless merging exposes readout sensitivity

A stage score can change even when the transformation discards no information. On NAVI, Qwen3.8 scores 0.3785 at the tower stage, 0.1130 after merging, and 0.3973 after projection. Merging concatenates neighboring tokens; it changes the vectors compared by cosine matching and their spatial sampling. The drop therefore cannot by itself establish information loss. Kimi K2.6 and

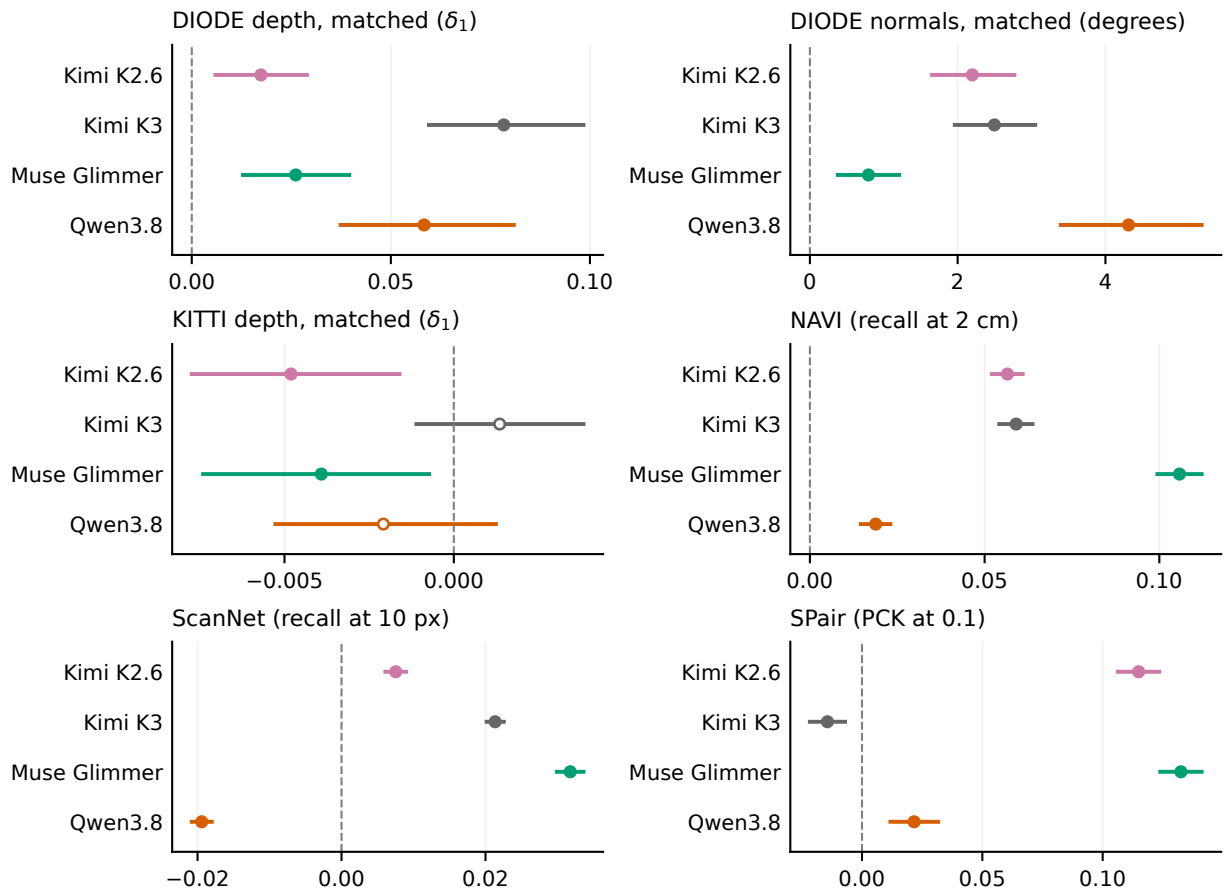


Figure 3: Projected features versus final tower features. Positive favors projected; for normals the difference is tower error minus projected error. DIODE and KITTI panels use matched readouts; correspondence is training-free. Bars are paired 95% intervals, with open markers where they cross zero. Each panel has its own metric and scale. Appendix D includes both geometry arms and exact values.

Muse Glimmer show the same drop-and-recovery pattern.

This control demonstrates sensitivity to representation format. It does not provide a numerical bound on the portion of a projector effect attributable to that sensitivity. Likewise, a trained depth probe can recover more from projected features without the projector adding new image information. Throughout, the measured outcome is performance under a particular readout and training procedure.

5 Implications for tower selection

The practical use of these results is to narrow candidates and extraction points for a specified task. SigLIP2 has the highest low-label recognition point estimate; DINOv2 has the highest geometry and NAVI estimates, often at intermediate layers for geometry. These descriptive patterns suggest candidates for validation, not an inferential ranking. At full labels, the Muse Glimmer–SigLIP2 recognition difference is unresolved, and the native-grid difference limits the interpretation of DINOv2 versus Qwen3.8 depth.

Tower and layer should be evaluated together. Their optima differ by task, and an intermediate

feature may be poorly matched to a projector trained on final-layer output. We have not tested reconnecting those layers to a language model. Nor do DIODE gains justify selecting a connector for another domain: KITTI and correspondence show that its measured effect can change sign. A deployment decision still requires the intended readout, data distribution, and downstream training recipe.

6 Limitations

The attention recognition results depend on 4×4 pooling whose full-token control is inconclusive. This affects both cross-tower comparisons and layer-curve shape; sharing a pooling operation does not guarantee it distorts all layers equally. Mean-readout agreement was established on two towers, not the entire roster. Geometry always uses full patch tokens.

DIODE and KITTI have only 115 and 150 test images, and DIODE readouts train on just 541 images. Equal training-set size does not imply equal underfitting across models. Bootstrap intervals resample images or pairs, not independent scenes, objects, or new training runs; correlations among units may make them overconfident. The selected headline comparisons are not a family-wide significance test. Layer peaks selected from test scores are descriptive and require separate validation before operational use.

PCA controls head width but can affect representations differently, and the full matched geometry matrix has not been rerun with deterministic PCA. Qwen3.8's 28×28 native grid differs from the other towers' 32×32 grids. A common-grid sensitivity experiment remains missing; neither PCA nor resizing correspondence maps establishes identical original spatial information. Kimi K3 also lacks parent forward-output and projector-weight parity checks, weakening confidence in its extracted pipeline.

Only rectangular occlusion was measured. The results do not establish robustness to blur, scale, lighting, or other distribution shifts. Finally, frozen probing evaluates access under our readouts. No language model is evaluated or adapted, and we do not test whether the observed stage orderings survive multimodal training.

7 Conclusion

Vision-tower selection depends on the capability, label budget, and extraction point. This six-tower study finds frequent intermediate-layer geometry peaks and consistent DIODE probe gains after native connectors, alongside reversals on KITTI and two correspondence comparisons. The useful next tests are domain-specific layer validation, a stable full-token recognition control, common-grid geometry comparisons, and downstream VLM evaluation of the selected representations.

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A Protocol and implementation details

A.1 Models and extraction

The standalone references are DINOv2-L [18] and SigLIP2-so400m-patch14-384 [23]. The extracted pipelines come from Kimi K2.6, Kimi K3, Muse Glimmer 30B, and Qwen3.8-27B [12, 15, 17, 19]. The repository adapters record their standalone checkpoint identifiers. The Qwen result filenames retain the internal slug `qwen3_5`; their recorded model ID identifies the Qwen3.8 checkpoint evaluated here. Likewise, `moonvit_v2` identifies the Kimi K3 tower. Display names in this paper follow the recorded models.

At 448×448, patch-14 towers produce 1,024 tokens and Qwen3.8 produces 784. Merging concatenates each 2×2 block, reducing these counts to 256 and 196 while quadrupling feature width. The merged widths are 4,608 for Kimi K2.6 and Qwen3.8, 4,096 for Kimi K3, and 6,144 for Muse Glimmer. Native normalization and packing are preserved; SigLIP2's position grid is interpolated from its configured resolution. Qwen tower tokens are restored from merge-block order to raster order, and intermediate Muse Glimmer tokens from attention-window order to raster order, before spatial scoring.

Extraction tests check exact parent weight equality for 335 packaged Kimi K2.6 tensors, 809 Muse Glimmer tensors, and 333 Qwen3.8 tensors, with parent stage-output comparisons. Kimi K3 has equality checks for its 165 tower tensors. Its three exported projector tensors are checked for compatible shapes, not equality to a separately loaded parent projector; there is no parent forward-output comparison. These different validation levels apply to every result that uses the respective adapter. Cache-level lossless-merge checks cover Kimi K2.6 and Kimi K3; losslessness for the other two is inferred from the implemented architecture.

A.2 Datasets and spatial alignment

ImageNet-100 uses the 13,000-image `validation` split of `ilee0022/ImageNet100` [9], exported by `scripts/export_imagenet100.py` and split with a fixed seed into 9,100/1,950/1,950 training/validation/test images. We identify the exact export because ImageNet-100 does not designate a unique class list or split. The four label budgets subsample only the training split and retain every class. Pooling to 16 tokens reduces recognition cache storage; the full-token validation is described in Appendix B.

DIODE uses 771 validation-release images with normals, comprising 325 indoor and 446 outdoor images. The 541/115/115 split is fixed. Native 768×1024 images are center-cropped consistently with the tower input; depth and normal targets retain their native resolution within that crop. The observed depth range sets the head's bin ceiling to 299.83 m. A Probe3D-style multiscale decoder [6, 20] reads one layer, rather than combining four encoder layers. We train for 10 epochs with AdamW [13], batch size eight. The records also contain δ_2, δ_3, RMSE, and scene-specific metrics. This single-layer, DIODE-based protocol differs from the original Probe3D evaluation.

KITTI uses the selected-validation archive of the depth-completion dataset, partitioned 700/150/150. Readouts receive RGB features only; sparse depth provides targets. Unsigned 16-bit depth is decoded in units of 1/256 m, zeros remain invalid, and invalid pixels are masked in loss and evaluation. Mean valid coverage is 17.08%, and the recorded maximum depth is 85.766 m. All towers use final tower features, with projected features additionally evaluated for the four connectors, in the matched arm.

NAVI uses 555 wild-set pairs with relative rotations in $(0, 120]$ degrees. ScanNet uses 1,500 pairs from the SuperGlue test split [21]. SPair-71k samples up to 200 pairs in each of 18 classes, for 3,600 pairs; the projected-versus-tower intervals use 3,408 pairs with finite scores at both stages. NAVI

and ScanNet features are bicubically resized to 64×64 before cosine nearest-neighbor matching; Lowe's ratio [14] ranks up to 1,000 retained matches. ScanNet RGB is aligned to the depth/intrinsics frame before cropping. SPair bilinearly samples source features at annotated keypoints, then selects target grid centers by cosine argmax. It uses each stage's own grid: 32×32 or 28×28 at the tower, and 16×16 or 14×14 after merging/projection. It uses neither Lowe ranking nor a top-1,000 cutoff. The SPair records' `evaluation_side: 64` field is unused by that scoring path. These protocols are adaptations of Probe3D. SPair aggregate levels are class-macro averages where labeled; bootstrap contrasts average paired image-pair differences and use a different weighting.

Occlusion uses a fixed 2,000-image ImageNet-100 subset, with 300 held-out test images. The reducer and attention readout are fitted on clean training features and frozen for all masking conditions. Records include requested and actual occluded area, rectangle coordinates, and fill value. Object scale and blur were planned but not measured.

A.3 Readout capacity and learning rates

Mean recognition averages tokens before a linear classifier. Attention recognition uses a learned query, a 512-wide multihead projection, layer normalization, and a classifier. The matched PCA arm fits the reduction on each cell's training features only, then freezes it. Parameter counts after reduction are 51,300 for mean recognition, 1,627,748 for attention recognition, 1,444,608 for depth, and 1,314,819 for normals. Raw heads grow with feature width. Matching parameter count does not ensure equal optimization difficulty or equal representation distortion.

Recognition searches $\{10^{-5}, 3 \cdot 10^{-5}, 10^{-4}, 3 \cdot 10^{-4}, 10^{-3}, 3 \cdot 10^{-3}, 10^{-2}, 3 \cdot 10^{-2}, 10^{-1}, 3 \cdot 10^{-1}, 1\}$; geometry searches $\{10^{-4}, 3{\cdot}10^{-4}, 10^{-3}, 3{\cdot}10^{-3}, 10^{-2}, 3{\cdot}10^{-2}\}$. The selected rate maximizes validation accuracy or depth δ_1, or minimizes validation normal error, as appropriate. Three retraining seeds provide the reported mean and population standard deviation. Paired image bootstrap intervals [5] average over those seeds before resampling, preserving the pairing between the two extraction points on each image.

B Protocol corrections and pooling control

The initial recognition grid began at $3 \cdot 10^{-4}$. Of 224 cells, 107 chose that floor: 53 of 56 matched-attention cells and 54 of 56 raw-attention cells, with no mean cells choosing it. The original run also had three upper-limit selections, for 110 boundary selections overall. After grid expansion and deterministic PCA correction, three selections remain at a boundary, all DINOv2 raw-mean cells at the upper limit. The experiment now records the complete validation search rather than only the selected rate.

Separately, randomized PCA originally depended on process RNG history. The reducer now seeds its randomized SVD inside a forked random-number context, leaving readout RNG state unchanged. All 112 matched recognition cells were rerun. Compared with the previous randomized matched run, 18 selected another learning rate; mean absolute top-1 changes were 0.00270 for attention and 0.00310 for mean pooling. The largest changes were -0.0110 for DINOv2 attention and $+0.0156$ for SigLIP2 mean pooling.

The original Muse Glimmer matched-attention merge-to-projector difference was -0.0075 top-1; the corrected result is $+0.0042$. Both learning-rate selection and PCA changed, so this contrast cannot isolate either correction's contribution. It also should not be described as a reversed paired significance test: the original claim used seed spread. The corrected attention stage changes are small and do not establish a general recognition gain or loss from projection.

The original matched geometry matrix cannot be reconstructed exactly because the PCA draws were not saved. Headline geometry comparisons were therefore retrained under the deterministic reducer; their point estimates can differ from the descriptive matrix. Recognition bootstrap comparisons also use separately retrained runs and can differ slightly from matrix or label-budget means. A bootstrap difference must be read from its own result record, not reconstructed by subtracting rounded entries in Table 2.

Pooling validation compares SigLIP2 and Muse Glimmer on a 1,500-image subset cached with both full tokens and 4×4 pooling. It covers both readouts and capacity arms. Mean readout rankings and curves agree, with raw means matching to four decimals. Six of sixteen full-token attention cells collapse at intermediate layers, preventing validation of the pooled attention curves. Diagnostics vary dataset size and training length on three unstable cells. With 5,000 images and 20 epochs, SigLIP2 matched at relative depth 0.630 scores 0.6796 ± 0.1551; extending training to 100 epochs stabilizes the two matched cells but leaves SigLIP2 raw at 0.7227 ± 0.0275. No tested configuration meets the stability target across all three cells. The records support an inconclusive attention-pooling control, not proof that pooling is harmless or that it causes the gap.

C Geometry layer peaks

Table 3 reports every geometry curve's best observed test score and its cost at the final layer. For depth the cost is peak minus final δ_1; for normals it is final minus minimum angular error. These maxima and minima are selected from the test sweep, not a separate validation selection. The matched arm uses the original randomized-PCA matrix. The single final-layer geometry peak is DINOv2 raw depth.

Table 3: Descriptive geometry peaks across both capacity arms. Depth scores and costs are δ_1; normal scores and costs are degrees.

Tower	Task	Arm	Peak depth	Peak score	Final-layer cost
DINOv2	Depth	matched	0.875	0.7015	0.0256
SigLIP2	Depth	matched	0.630	0.6702	0.0764
Muse Glimmer	Depth	matched	0.760	0.6588	0.1227
Kimi K2.6	Depth	matched	0.519	0.6595	0.0593
Qwen3.8	Depth	matched	0.630	0.6736	0.1422
Kimi K3	Depth	matched	0.630	0.6546	0.1167
DINOv2	Normals	matched	0.750	18.8508	4.3426
SigLIP2	Normals	matched	0.370	23.9039	7.1108
Muse Glimmer	Normals	matched	0.620	24.3014	7.2328
Kimi K2.6	Normals	matched	0.370	25.5109	5.1786
Qwen3.8	Normals	matched	0.370	24.2386	10.2153
Kimi K3	Normals	matched	0.370	27.0277	4.7317
DINOv2	Depth	raw	1.000	0.6746	0.0000
SigLIP2	Depth	raw	0.519	0.6366	0.0546
Muse Glimmer	Depth	raw	0.620	0.6035	0.1946
Kimi K2.6	Depth	raw	0.741	0.5089	0.0446
Qwen3.8	Depth	raw	0.630	0.6439	0.3001
Kimi K3	Depth	raw	0.630	0.5440	0.0878
DINOv2	Normals	raw	0.750	19.4649	2.7510
SigLIP2	Normals	raw	0.370	25.1602	5.4407
Muse Glimmer	Normals	raw	0.620	22.7825	11.0423
Kimi K2.6	Normals	raw	0.889	30.3440	1.7705
Qwen3.8	Normals	raw	0.519	24.6806	9.0479
Kimi K3	Normals	raw	0.630	27.9684	5.3326

D Paired comparisons

All tables in this section are generated from the paired result JSONs. Intervals are pointwise 95% bootstrap intervals with 10,000 resamples. Positive advantages favor the first representation. For normal error, the sign is reversed so positive means lower error. Each table uses its stated evaluation unit and capacity setting.

D.1 DIODE

The following two tables cover all 16 projected-versus-final-tower comparisons on 115 images. Depth uses δ_1 and normals use degrees of error reduction. DINOv2 and SigLIP2 have no native projected stage and are not included.

Table 4: DIODE projected-versus-tower differences, matched arm.

Connector	Task	Difference	95% interval
Kimi K2.6	Depth	+0.0174	[+0.0060, +0.0289]
Kimi K3	Depth	+0.0784	[+0.0596, +0.0984]
Muse Glimmer	Depth	+0.0261	[+0.0129, +0.0395]
Qwen3.8	Depth	+0.0584	[+0.0374, +0.0809]
Kimi K2.6	Normals	+2.1980	[+1.6538, +2.7685]
Kimi K3	Normals	+2.4968	[+1.9616, +3.0492]
Muse Glimmer	Normals	+0.7924	[+0.3770, +1.2065]
Qwen3.8	Normals	+4.3130	[+3.3956, +5.3018]

Table 5: DIODE projected-versus-tower differences, raw arm.

Connector	Task	Difference	95% interval
Kimi K2.6	Depth	+0.0196	[−0.0003, +0.0402]
Kimi K3	Depth	+0.0270	[+0.0113, +0.0427]
Muse Glimmer	Depth	+0.0927	[+0.0624, +0.1237]
Qwen3.8	Depth	+0.1601	[+0.1211, +0.1977]
Kimi K2.6	Normals	+3.1126	[+2.4215, +3.8228]
Kimi K3	Normals	+2.8269	[+2.1506, +3.5442]
Muse Glimmer	Normals	+2.8932	[+2.3767, +3.4432]
Qwen3.8	Normals	+4.4432	[+3.5076, +5.4427]

Scene-specific intervals for the matched arm use 51 indoor and 64 outdoor images. Fifteen of these sixteen intervals exclude zero; Kimi K2.6 outdoor depth is unresolved.

Table 6: DIODE projected-versus-tower differences by scene, matched arm.

Connector	Task	Scene	Difference	95% interval
Kimi K2.6	Depth	Indoor	+0.0325	[+0.0145, +0.0506]
Kimi K2.6	Depth	Outdoor	+0.0053	[−0.0085, +0.0198]
Kimi K3	Depth	Indoor	+0.1170	[+0.0813, +0.1541]
Kimi K3	Depth	Outdoor	+0.0476	[+0.0333, +0.0643]
Muse Glimmer	Depth	Indoor	+0.0336	[+0.0099, +0.0584]
Muse Glimmer	Depth	Outdoor	+0.0201	[+0.0064, +0.0347]
Qwen3.8	Depth	Indoor	+0.0499	[+0.0153, +0.0858]
Qwen3.8	Depth	Outdoor	+0.0651	[+0.0393, +0.0930]
Kimi K2.6	Normals	Indoor	+3.2144	[+2.2093, +4.2502]
Kimi K2.6	Normals	Outdoor	+1.3880	[+0.8998, +1.9298]
Kimi K3	Normals	Indoor	+3.5815	[+2.6695, +4.5369]
Kimi K3	Normals	Outdoor	+1.6324	[+1.1034, +2.2109]
Muse Glimmer	Normals	Indoor	+1.1128	[+0.3023, +1.9190]
Muse Glimmer	Normals	Outdoor	+0.5372	[+0.1624, +0.9161]
Qwen3.8	Normals	Indoor	+6.8576	[+5.2091, +8.6752]
Qwen3.8	Normals	Outdoor	+2.2853	[+1.7216, +2.8849]

D.2 KITTI and correspondence

KITTI differences use δ_1, matched readouts, and 150 test images. Correspondence has no trained head or capacity arm: NAVI uses recall at 2 cm over 555 pairs, ScanNet recall at 10 pixels over 1,500 pairs, and SPair PCK at 0.1 over 3,408 finite pairs. SPair differences weight pairs equally, rather than weighting classes equally.

Table 7: KITTI projected-versus-tower differences.

Connector	Difference	95% interval
Kimi K2.6	−0.0048	[−0.0077, −0.0016]
Kimi K3	+0.0014	[−0.0011, +0.0038]
Muse Glimmer	−0.0039	[−0.0074, −0.0007]
Qwen3.8	−0.0021	[−0.0053, +0.0012]

Table 8: NAVI projected-versus-tower differences.

Connector	Difference	95% interval
Kimi K2.6	+0.0565	[+0.0521, +0.0609]
Kimi K3	+0.0590	[+0.0542, +0.0637]
Muse Glimmer	+0.1058	[+0.0995, +0.1121]
Qwen3.8	+0.0188	[+0.0146, +0.0230]

Table 9: ScanNet projected-versus-tower differences.

Connector	Difference	95% interval
Kimi K2.6	+0.0075	[+0.0061, +0.0090]
Kimi K3	+0.0213	[+0.0202, +0.0225]
Muse Glimmer	+0.0318	[+0.0299, +0.0336]
Qwen3.8	−0.0194	[−0.0208, −0.0180]

Table 10: SPair-71k projected-versus-tower differences.

Connector	Difference	95% interval
Kimi K2.6	+0.1150	[+0.1063, +0.1235]
Kimi K3	−0.0144	[−0.0216, −0.0070]
Muse Glimmer	+0.1325	[+0.1239, +0.1411]
Qwen3.8	+0.0216	[+0.0118, +0.0316]

D.3 Recognition and cross-tower geometry

Recognition comparisons include both between-tower and within-Qwen layer contrasts, all paired over 1,950 test images. Each representation is labeled by its relative depth. The matched attention Muse Glimmer–DINOv2 contrast and both raw Qwen earlier-layer contrasts exclude zero. The remaining recognition intervals cross zero.

Table 11: Recognition differences, first minus second top-1 accuracy.

First / second	Readout	Arm	Difference	95% interval
Muse Glimmer 1.000 / DINOv2 1.000	attention	matched	+0.0125	[+0.0034, +0.0215]
Muse Glimmer 1.000 / SigLIP2 1.000	attention	matched	+0.0041	[−0.0036, +0.0121]
Qwen3.8 0.741 / Qwen3.8 1.000	attention	matched	+0.0017	[−0.0063, +0.0096]
SigLIP2 1.000 / DINOv2 1.000	attention	matched	+0.0084	[−0.0009, +0.0178]
Muse Glimmer 1.000 / SigLIP2 1.000	attention	raw	−0.0017	[−0.0097, +0.0063]
Qwen3.8 0.889 / Qwen3.8 1.000	attention	raw	+0.0128	[+0.0053, +0.0203]
Muse Glimmer 1.000 / SigLIP2 1.000	mean	matched	−0.0039	[−0.0130, +0.0053]
Qwen3.8 0.889 / Qwen3.8 1.000	mean	matched	+0.0015	[−0.0056, +0.0085]
Muse Glimmer 1.000 / SigLIP2 1.000	mean	raw	−0.0010	[−0.0104, +0.0085]
Qwen3.8 0.889 / Qwen3.8 1.000	mean	raw	+0.0156	[+0.0053, +0.0263]

The two cross-tower geometry comparisons rerun test-sweep-selected layers with deterministic PCA: depth at relative depths 0.875 for DINOv2 and 0.630 for Qwen3.8, and normals at 0.750 for DINOv2 and 0.370 for SigLIP2. Layer selection has no independent validation. DINOv2–Qwen3.8 depth compares different native grids; DINOv2–SigLIP2 normals uses the same token count. The rerun depth levels are 0.6988 and 0.6697; normal errors are 18.8204 and 23.8575 degrees. They need not equal the original sweep's values.

Table 12: Cross-tower geometry advantages under deterministic matched reruns.

First / second	Task	Advantage	95% interval
DINOv2 / Qwen3.8	Depth	+0.0292	[+0.0099, +0.0484]
DINOv2 / SigLIP2	Normals	+5.0370	[+4.3038, +5.8164]

E Occlusion and correspondence levels

Figure 4 shows absolute occluded-image accuracy and NAVI scores through the connector. Its occlusion panel is not a clean-minus-perturbed difference. Within-stage clean-versus-occluded intervals resolve a loss at 10% masking for all multimodal stages. DINOv2's tower is unresolved at that level and first resolves at 20%. The apparent cross-tower retention groups have not been tested with paired retention intervals. No connector stage is uniformly more resistant to occlusion.

The NAVI panel illustrates the scoring effect of merging. Qwen3.8, Kimi K2.6, and Muse Glimmer fall after concatenation and recover after projection. This demonstrates format sensitivity of direct cosine matching; the numerical size of that fall is not a bound on information loss or recovery in the adjacent stage.

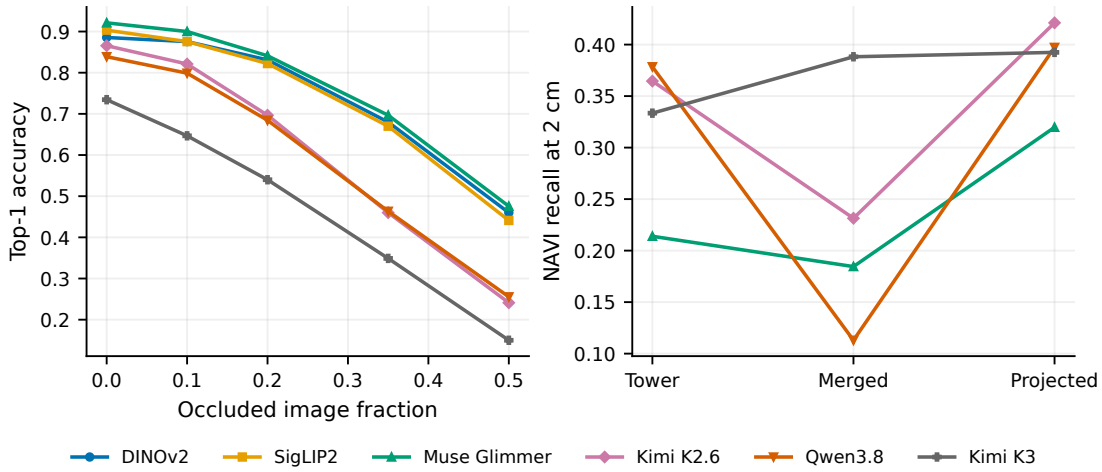


Figure 4: Left: absolute top-1 accuracy under occlusion at the tower stage, matched attention readout. Right: training-free NAVI recall at the three native stages for multimodal towers. Curves are descriptive; uncertainty is not plotted.

Table 13: Final-tower correspondence levels. SPair is a macro average over 18 classes; its paired intervals above average over finite image pairs instead.

Tower	NAVI	ScanNet	SPair (macro)
DINOv2	0.5389	0.0896	0.5547
SigLIP2	0.4013	0.0584	0.3982
Muse Glimmer	0.2139	0.0208	0.0909
Kimi K2.6	0.3646	0.0547	0.1782
Qwen3.8	0.3785	0.0834	0.3081
Kimi K3	0.3334	0.0258	0.2760

F Result provenance and reproduction

Source and results: `https://github.com/immanuel-peter/vision-tower-bench`. Build instructions are in `paper/README.md`. `paper/build_assets.py` generates numerical assets from recorded JSONs, with input hashes in `paper/asset-sources.json`. `results/README.md` indexes the experiments and corrections.